

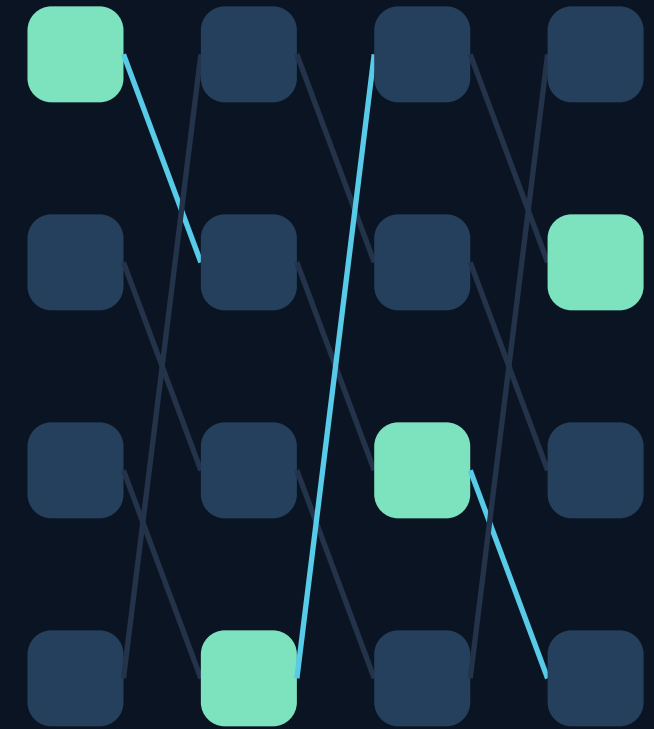
Sleeping Machines

AI that learns what to compute, when to compute it, and what to remember.

A trainable model family and computing substrate built around time, local memory and selective work.

Tero Keski-Valkama | Sole founder

Raising €3M to scale research and engineering
Research-stage deep tech · 4 October 2026



Capacity beyond activity

Model capacity and the cost of using it must scale differently.

The opportunity: preserve useful intelligence while reducing the work paid for each prediction.

MODEL CAPACITY

Know more

Larger models and richer memory can improve capability. Accessing that capacity still costs computation and data movement.

DEPLOYMENT ECONOMICS

Do less work

Serving cost, latency and memory determine whether capability can be deployed in constrained environments.

OUR APPROACH

Select + retain

Learn which local computations to activate; retain local state and train the alternatives that were not selected.

We are developing a model family and execution substrate around this separation.

One model family across today's architectural landscape.

Stateful temporal programs connect representation, memory, timing and learning.

Recurrent / SSM models

Persistent state and selective dynamics.
Add addressed temporal programs.

Attention / Transformers

Content-addressed aggregation.
Include rich local interaction.

Sparse experts / MoE

Capacity beyond selected expert work.
Add private facts and timing.

SLEEPING MACHINES / STATEFUL TEMPORAL PROGRAMS

One content / time / address / state interface; compatible choices from units to complete systems.
Integrated target: computational time, selective state, small messages, separate keys/values and unrealized credit.

Statistics / retrieval

Counts and direct historical evidence.
Combine with learned abstractions.

Adaptive memory / TTT

Learning in the deployed model.
Use local, causal update policies.

Event / temporal systems

Local arrivals, timers and causal order.
Add vector messages and route credit.

Containment needs actual operators and contracts. Learned parity, lower total cost and physical energy require evidence.

Choose the right computation in each region.

One family can mix selective temporal programs with rich interaction and causal memory.

01 / UNITS

Local programs

Choose state, flow, writes, small messages, keys and reception. Time can perform computation.



02 / LAYERS

Adaptive work

Select one or many messages. Add dense interaction where needed; preserve lean regions elsewhere.



03 / COMPOSITIONS

Shared evidence

Link learned vectors, counts and historical memory. Explicit paths let dense and sparse inputs inform one query.



04 / SYSTEMS

Same learner

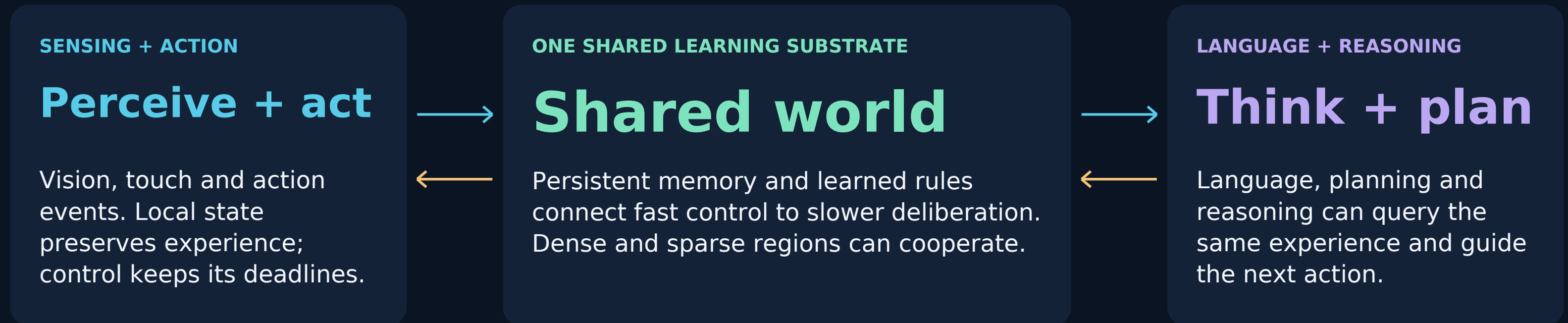
Support streamed or batched updates and native TTT. Broader structural design is a further learning ambition.

Counterfactual credit: learn from selected and unrealized routes

No required global periodic tick. Joint integration, measured TTT gains, morphing and hardware energy remain milestones.

A substrate for intelligence that learns across mind and body.

The ambition: connect sensing, action, language and reasoning in one trainable model, across their natural timescales.



Hypothesis: physical experience teaches reusable abstractions; reasoning improves action. Online learning keeps both adapting.

Upside: persistent generalist agents, from datacenter software to embodied intelligence on mobile robots.

First proof: held-out transfer in both directions, with retained skills and matched data/work. AGI is an ambition, not a demonstrated capability.

Messages race. Selected memories update. Alternatives learn.

A simple event loop connects representation, timing and sparse computation.

01 / ENCODE

Signal + delay

A learned function produces a message and its arrival time from input and local memory.



02 / SELECT

Temporal race

Arrival times select a receiver. Separate address keys and delivered values support sparse execution.



03 / REMEMBER

Local state

Update the selected memory and propagate a new message into deeper computation.



04 / LEARN

Alternatives

Use errors to improve message content and timing, including paths that were not selected.

Counterfactual credit: learn from selected and unrealized routes

Implemented language core: hard selection and persistent local state in a synchronous emulator.

The language experiments establish three useful facts.

10M-character text8 training budget; held-out prediction error. Lower bits per character is better.

IT CAN LEARN REAL TEXT

2.162 bits/char

Best model reaches the saved recurrent baseline's quality (2.171). It uses 2.89× estimated training work.

ALTERNATIVE CREDIT HELPS

−0.135 bits/char

In the same small model, learning from unchosen paths improves prediction error; estimated training work rises 0.30%.

CAPACITY AND SELECTED WORK

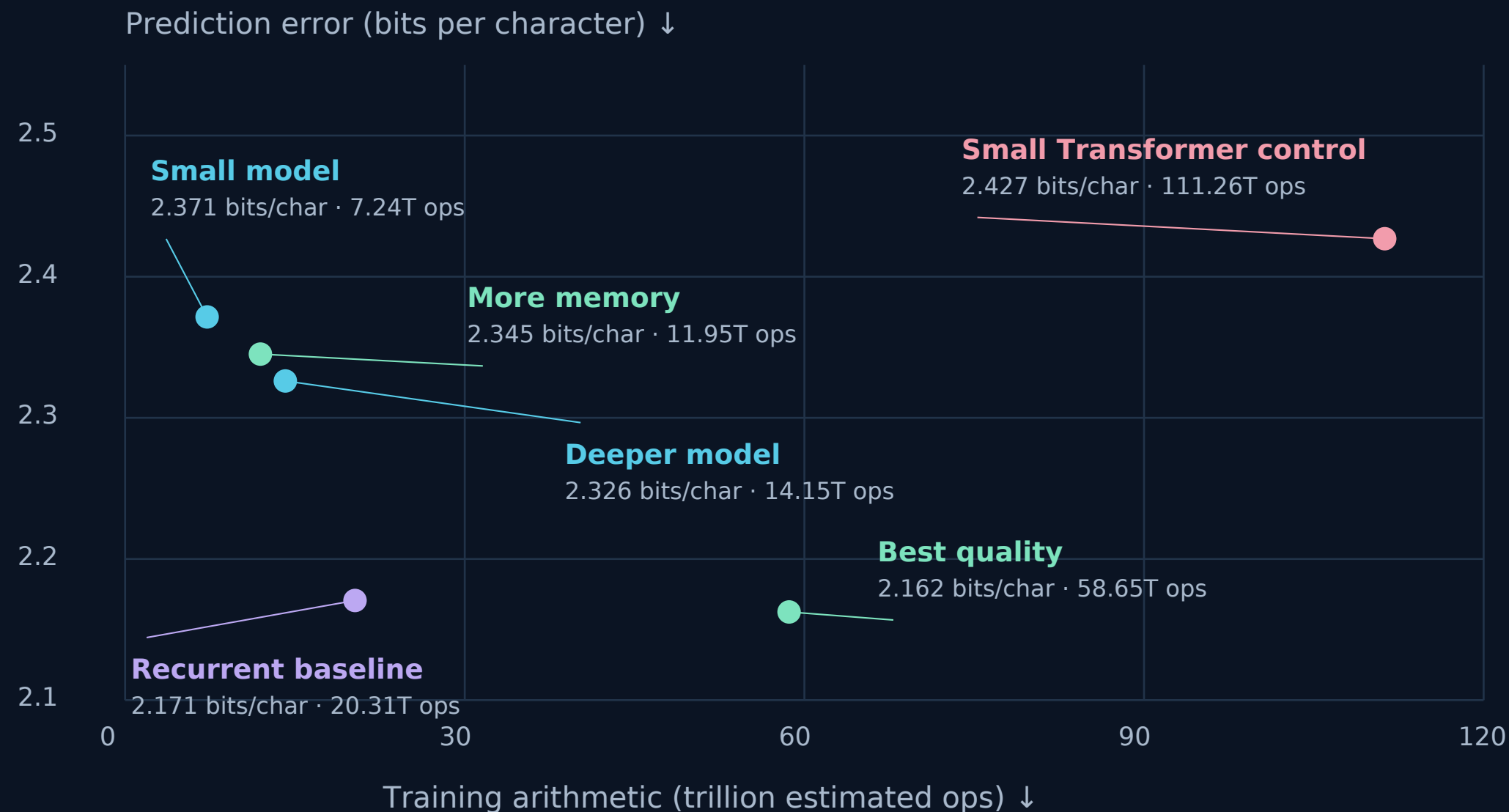
16 → 32 slots

More available memory improves prediction while retaining eight selected writes per input. Key scoring and learning costs also grow.

Exploratory single-seed results. The next proof is replicated quality plus a measured system-cost advantage.

A language-model foothold for a new architecture.

Held-out prediction error versus estimated training arithmetic; 10M characters, one pass.



QUALITY CHECK

2.162

Near recurrent baseline (2.171); 2.89× training work.

TRAINING ARITHMETIC

15.4× less

Small model versus saved Transformer control; better prediction error.

Saved one-pass controls; same test region and window length. Training order and work conventions differ. One seed. Arithmetic estimates are not service cost.

Start with low-batch language inference on existing hardware.

Product hypothesis for local and on-premise text applications; no customer interest reported yet.

WHO WOULD EVALUATE IT

Platform teams

Teams deploying small language models under latency, memory or cost constraints. Demand validation is a next step.

WHAT WE WOULD SUPPLY

Model + runtime

A trained model, sparse inference runtime and a reproducible quality-and-cost evaluation on the intended workload.

WHY THEY WOULD ADOPT

Verified savings

Adoption depends on equivalent useful quality, service reliability and a measured reduction in complete deployment cost.

Commercial gate: target $\geq 20\%$ full-system cost reduction at matched quality on one selected workload.

Software establishes the advantage. Hardware can amplify it.

One research program, with staged routes to deployment.

01 / MODELS + RUNTIME

First product

Validate the language workload on existing hardware. License a model/runtime with auditable deployment economics.

02 / ACCELERATOR INTEGRATION

Co-design

Optimize supported hardware, compilers and memory access. Pursue joint evaluations and platform-specific licensing.

03 / EVENT-DRIVEN SILICON

Clockless path

Translate learned timing and local state into hardware feasibility work. Expand to mobile, robotics and persistent agents.

The €3M round funds scale, execution and feasibility; later silicon investment follows measured proof.

A useful alternative could strengthen a hardware ecosystem.

Potential evaluation and licensing routes; no partner interest or relationship is claimed.

AMD

Model + platform

Its ~\$665M Silo AI acquisition and Liquid AI collaboration show the value of model/software capability on AMD hardware.

INTEL

Event execution

Hala Point and Loihi 2 provide a relevant research environment for expressive event models and continuous learning.

GOOGLE

Full-stack design

TPU model-hardware co-design creates a demanding evaluation venue for quality, memory and execution improvements.

Partner proposition: demonstrate a differentiated workload advantage that improves platform adoption.

Price the runtime against verified customer value.

Illustrative annual license economics; assumptions to test with an evaluation partner.

Per-customer assumption	Calculation	Annual amount
Eligible workload cost	Existing deployment cost we can actually influence	€1.00M
Gross savings	20% complete-cost reduction at matched quality	€200k
Runtime license	20% of verified gross savings	€40k

Customer retains €160k/year before migration and other costs. At the same assumed price, 25 deployments would generate €1M annual license revenue.

Validate pricing through matched quality, reliable operation and measured customer benefit.

AI research and software architecture in one founder.

Tero Keski-Valkama · Sole founder · Public career and invention record.

Tero Keski-Valkama

Sole founder

Experience spans applied ML, multimodal foundation models and software architecture. The round adds specialist capacity around the founder.

2012-2018

Cybercom · Software Architect

Historical role in public professional profile.

2018-2022

HERE · Lead ML & AI Engineer

Self-reported research and applied ML leadership.

2022-2025

Alloy.ai → Kaiko.ai

Staff engineering; multimodal foundation-model work.

2023 publication

EP4148389A2 · Named inventor

Maps from geospatial observations; applicant HERE.

Historical roles are self-reported; inventorship is documented. Planned hiring: learning research, runtime engineering, evaluation and measurement.

Build defensibility across the model and its execution.

The strongest potential asset is an integrated stack that is difficult to reproduce and deploy.

MODEL + LEARNING

Research know-how

Timing, state and alternative-credit methods, supported by retained ablations and numerical contracts.

RUNTIME + HARDWARE

Execution know-how

Sparse kernels, memory management, measurement and hardware mapping can reinforce the model advantage.

RIGHTS + INTEGRATION

Company assets

Establish contribution rights and licensing, then develop tooling and partner-specific deployment expertise.

Private development supports controlled disclosure. IP ownership and commercial integrations remain diligence items.

Three gates determine whether this becomes a business.

Each gate has a falsifiable result and a consequence for further spending.

01 / QUALITY AT SCALE

Replicate

At least three seeds, fixed causal tests and tuned controls. Compare quality and complete work; keep failures beside the best result.

02 / COMPLETE ECONOMICS

Measure

Pass trained sparse parity. Measure total serving cost, tail latency, memory and energy at useful quality on one fixed workload.

03 / ADOPTION

Evaluate

Identify a use case and an evaluator willing to test it. Confirm useful quality, integration effort and willingness to pay.

If a gate fails, diagnose or narrow the workload before committing more compute or new silicon.

The next 18 months produce evidence investors can underwrite.

Proposed milestones measured from funding; spending advances with completed gates.

MONTHS 0-3

Team + proofs

Hire the initial team.
Establish company/IP rights. Complete trained sparse execution checks and larger-data, multi-seed evaluation.

MONTHS 3-6

One workload

Run tuned controls and measure total cost, latency, memory and energy. Select a workload only if quality and economics hold.

MONTHS 6-12

External trial

Seek independent reproduction and prospective customer evaluations. Test pricing, integration effort and willingness to pay.

MONTHS 12-18

Product + scale

Turn a validated workload into a deployable runtime. Expand models and assess hardware mapping for the next financing stage.

Next-round objective: replicated quality, a measured cost advantage and a credible evaluation pathway.

€3M funds an 18-month research and engineering program.

A proposed six-person average team, staged compute and hardware feasibility.



- Team: research + systems €1.35M | 45.0%
- Compute + independent replication €0.90M | 30.0%
- Hardware feasibility + measurement €0.25M | 8.3%
- IP, legal, operations €0.20M | 6.7%
- Contingency reserve €0.30M | 10.0%

€2.70M operating plan + €0.30M reserve · Planned operating burn averages €150k/month.
Budget assumptions: six average FTE including founder at €150k/year fully loaded; compute and hiring quotes remain to be obtained.

Raise €3M to establish a commercially valuable platform.

Proposed priced round; terms remain subject to company formation and investor diligence.

CAPITAL TO RAISE

€3M

18 months of research, systems engineering and staged validation.

PROPOSED PRE-MONEY

€50M

A premium research-stage proposal for the model/runtime/substrate opportunity.

INITIAL OWNERSHIP

5.66%

€53M post-money, before option-pool changes, preferences and later dilution.

The proposal prices a model/learning family, sparse runtime and future compute substrate. Adaptive communication and embodied learning widen the opportunity; €3M scales the team and decisive proof points.

Valuation sensitivities and conditional exit scenarios are in the appendix.

Build the model. Prove the economics. Open the hardware path.

€3M proposed raise · €50M proposed pre-money · Private investor discussion.

€3 million

A new architecture is becoming trainable.
This round tests whether its advantage scales.

- 01** Replicated language quality and learning mechanisms.
- 02** A deployable runtime with measured system economics.
- 03** External evaluation and a credible path to hardware.

Tero Keski-Valkama · Private investor diligence available by arrangement

The same model improves when unchosen paths receive credit.

Controlled small-model comparison; held-out error and estimated complete training work.

TEST BITS / CHARACTER

2.506 → 2.371

Timing-only credit versus added local alternative-value route credit.

ESTIMATED FITTING WORK

+0.30%

Hard forward selection and parameter count retained.

CHARACTER PERPLEXITY

−8.9%

Derived from the BPC difference; one controlled seed.

Same small model; hard selection retained

A deeper variant also improves by 0.130 bits per character. Both are single-seed comparisons; numerical details and model definitions follow.

A falsifiable architectural mechanism, with a small counted arithmetic increment.

More available state. The same eight selected writes.

Small model with alternative credit: available memory doubles; selected updates stay fixed.

Pool 2



16 slots · 512 value scalars
8 writes · 16 keys scored
2.371 bpc

Pool 4



32 slots · 1,024 value scalars
8 writes · 32 keys scored
2.345 bpc

WHAT SCALES

8 writes

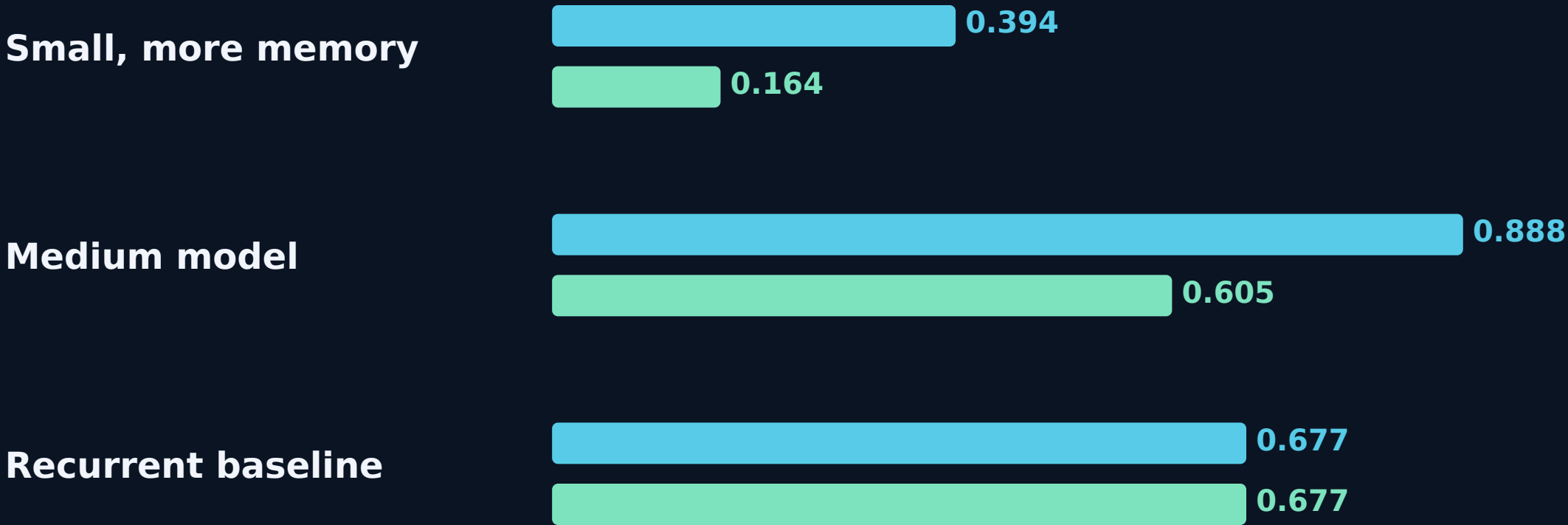
Winner-only arithmetic +0.66%.
Fitting work +65.0%. Keys and
value storage double.

Quality improves by 0.026 bpc; total capacity and learning costs remain explicitly charged.

The sparse path is implemented; system proof is next.

Estimated arithmetic per evaluated input position. Green bars show the implemented winner-only path.

Estimated MFLOPs per evaluated input position



STATUS

All keys remain scored. Actual trained FP32 parity and held-out rescore are pending. Traffic, setup, latency and joules must be measured.

Cyan: emulator / control forward Green: winner-only / same control

These are backend/shape estimates. Native evaluation pays overlapping-window warmup (~2 positions/target); LSTM recurrent evaluation differs. Not a service-cost win.

An adaptation signal exists. Its work is accounted for.

Predict-before-update experiment on a new text stream; separate from the latest model family.

PREDICT BEFORE UPDATE

3.191 → 3.096

8,191 new development targets; same persistent state, frozen versus full-backbone adaptation.

COSTED LEARNING

10.7× work

Total online processing work includes credit, backward and optimizer. Retention under drift and native asynchronous learning remain open.

Online-learning upside: persistent adaptive systems. Current result: one costed pilot.

One substrate can support new communication and learning patterns.

Application hypotheses with first proofs; no project codec, embodied-transfer or AGI result.

Opportunity	Architectural fit	First discriminating proof
Irregular inputs	Addressed events retain time and update persistent programs. Dense regions remain available.	Same causal data/queries; compare timestamp-aware controls and burst/idle resource use.
Asynchronous codecs	Learn what, where and when to send; decoder retains predictive state.	Actual bytes versus distortion/task quality, sender/receiver work and causal latency.
Memory meta-learning	Slow rules learn fast writes, retention, retrieval and scheduling from later utility.	Held-out delayed-use tasks; complete useful-return credit against immediate utility.
Embodied cognition	Shared sensing/action/language paths can exchange learned physical abstractions.	Motor-to-cognitive and reverse transfer measured separately; matched data/work and isolated-path controls.

The contribution is the complete construction.

Established alternatives already address parts of this problem; they remain essential controls.

Approach	Established strength	Our comparison obligation
Transformer / MoE	Strong quality; MoE separates available parameters from selected expert work.	Match quality/data and charge routing, memory traffic and training.
RNN / selective SSM	Persistent state and efficient sequence processing.	Test whether temporal races add useful quality or allocation.
Neuromorphic systems	Event-driven hardware and local learning precedents.	Prove expressive model quality, mapping and measured energy.
Sleeping Machines	Joint timing, sparse memory, deep representation and alternative credit.	Scale integrated mechanisms without replacing them with dense carriers.

Sparsity, recurrence and on-chip learning alone are not the novelty claim.

AMD's Silo AI deal supports the strategic logic.

Announced approximately \$665M all-cash acquisition in 2024; a mature-company precedent.

WHAT AMD ACQUIRED

Team + stack

AI scientists and engineers, enterprise solutions, model development and software capability on AMD hardware.

THE LUMI CONNECTION

Pororo / Viking

AMD highlighted multilingual Pororo and Viking models trained on AMD-powered LUMI, with university and supercomputer collaborators.

OUR LESSON

Prove leverage

Build a capability that improves a platform's economics and adoption. One small research project is not a direct pricing comparable.

Efficient models can matter to a chip ecosystem; training on its chips is one part of the value case.

The constraints reach from power grids to robot batteries.

Economic context for efficient intelligence; these figures are not our revenue market.

DATACENTERS

485 → 950

TWh/year, 2025 → 2030 central IEA projection. Power and data movement make system efficiency economically important.

MOBILE

5.8B

Unique mobile subscribers in GSMA's 2026 report. On-device inference creates a prospective route to broad distribution.

ROBOTICS

5M

Industrial robots operating in 2025. Sparse sensing, persistent state and adaptation are prospective use cases.

Select one measurable initial workload; expand only after quality and system gains hold.

Successful platform adoption can support very large outcomes.

Conditional commercial arithmetic illustrates the scale required; no forecast or market multiple claim.

Successful outcome assumption	Value capture assumption	Equity-value scenario
€10B/year eligible compute spend	20% savings × 20% capture = €400M/year; 25× revenue assumption	€10B
€50B/year eligible compute spend	20% savings × 25% capture = €2.5B/year; 20× revenue assumption	€50B

A cross-platform substrate could participate in cloud inference, persistent agents and embodied systems. Reaching these outcomes requires sustained quality, scale, margins and broad adoption.

The upside is large if the platform works and wins adoption. Its probability remains unmeasured.

The completed comparisons, using consistent resource units.

Frozen 10M-character, one-pass evidence. Model definitions are on the next slide.

Model	Test bpc	Params K	Fit total TF	Fit MF/pos	Emul MF/pos	Winner MF/pos
Small: timing-only credit	2.506	108.9	7.22	0.722	0.240	0.163
Small: alternative credit	2.371	108.9	7.24	0.725	0.240	0.163
Small: more memory	2.345	177.0	11.95	1.196	0.394	0.164
Deeper: alternative credit	2.326	210.0	14.15	1.416	0.463	0.310
Medium: alternative credit	2.183	422.5	26.79	2.680	0.888	0.605
Medium: more memory	2.179	689.8	44.07	4.410	1.458	0.607
Best quality: wider model	2.162	940.9	58.65	5.868	1.947	1.324
Recurrent baseline (LSTM)	2.171	338.4	20.31	2.032	0.677	0.677
Small Transformer control	2.427	1658.9	111.26	11.133	3.710	3.710

Fit MF/pos = per training position. Inference MF/pos = per evaluated input position. Native quality uses training evaluator; winner-only trained parity/rescore pending. Native overlap warmup adds work per scored target. Full 15-row CSV retains older models.

Plain-language labels map to reproducible model records.

Message width counts scalars; available memory counts local state slots.

Deck label	Width / layers	Memory slots	Record configuration
Small model	32 / 4	16	p32 / d4 / pool2
Small: more memory	32 / 4	32	p32 / d4 / pool4
Deeper model	32 / 8	32	p32 / d8 / pool2 / skip2
Medium model	64 / 4	16	p64 / d4 / pool2
Medium: more memory	64 / 4	32	p64 / d4 / pool4
Best quality	96 / 4	16	p96 / d4 / pool2
Saved controls	256 hidden width	Conventional architectures	LSTM-256; Transformer-256 × 2 layers

The comparison boundaries are part of the evidence.

Maintain the original records and disclose what a result actually measures.

Boundary	Current completed evidence	Remaining work
Data / evaluation	Frozen table: 10M fit. Report: completed 90M fit, 1.8573 BPC on test[95M:96M], T256.	Align modern controls, target coverage and fit work; full official 5M test remains open.
Learning / seeds	10M ledger: 1,220 updates, lr .004 cosine; one seed; state reset per fit segment.	Replication, robust recipe, depth/horizon and matched larger-data comparisons.
Sparse execution	All keys scored; selected values only in winner evaluator; small random FP64 equality fixtures.	Actual trained FP32 winner/state/cache equality and held-out rescore.
System economics	Arithmetic traces/formulas; persistent local state within windows.	Setup, warmup, traffic, memory, throughput, latency, joules and total cost.
Adaptive workload	Fixed scored queries; declare late/missing readout, feedback and fallback.	Charge attempted work; report coverage separately if abstention is the task.

Keep the failures beside the positive mechanism results.

A disciplined research program should revise interpretations when evidence changes.

Finding	Interpretation	Response
Uncredited D8 pool expansion worsens quality	More capacity needs learnable selection; it is not automatically useful.	Retain negative pair; test credited expansion with matched depth/seed.
Write-credit variants regress or diverge	A particular local estimator is unstable; not all route credit is equivalent.	Withdraw variants; preserve original result and derive corrected credit.
Best p96 costs more than LSTM	Small quality lead is not comparable-quality resource superiority.	Tune and scale; require complete quality-cost frontier.
Event-vision screening is mixed	Small coarsening gains do not clear every full-screen seed or strong RBF controls.	Avoid cross-domain supremacy claim; keep modality-specific gates.

Reverse-underwriting shows the assumptions behind a price.

This arithmetic is a sensitivity tool; the research does not estimate platform-success probabilities.

BULLISH PRICED ROUND

€50M

€3M raise → €53M post-money;
5.66% new-investor ownership.

REVERSE UNDERWRITING

6.74%

Required platform-success
probability in the stated €10B exit
scenario.

STRETCH SCENARIO

€100M

€100M pre-money requires 13.49%
in the same model; 2.91%
ownership.

$$V = p \times \text{exit equity} \times \text{future retention} / (1 + \text{discount rate})^{\text{years}}$$

Assumptions: €10B exit; 30% stake retention; 10 years; 15% discount; failure value zero.

Required odds are reverse-calculated conditions. The benchmarks do not estimate those probabilities.

The price depends on dilution, exit scale and time.

Required platform-success odds at 10 years and 15% discount; zero failure value.

Exit equity	Stake retention	Required p: €50M	Required p: €100M
€3B	10%	67.43%	134.9% (infeasible)
€3B	30%	22.48%	44.95%
€3B	50%	13.49%	26.97%
€10B	10%	20.23%	40.46%
€10B	30%	6.74%	13.49%
€10B	50%	4.05%	8.09%
€30B	10%	6.74%	13.49%
€30B	30%	2.25%	4.50%
€30B	50%	1.35%	2.70%

Sensitivity assumptions only. Probabilities above 100% are infeasible in this model. Preferences, option pools and intermediate financing cash flows are omitted.

A private evidence pack and primary external context.

Source labels in slide footers are clickable; companion files must travel with the PDF.

R1 **Private completed language evidence ledger (3 October 2026)**

Saved native fits and E64 controls. Original JSON records and SHA256 hashes are supplied in the private diligence pack.

R2 **Private online-learning experiment, 8,191 targets**

Predict-before-update pilot: quality and complete processing-operation estimates; one checkpoint, stream and learning rate.

R3 **Architecture, numerical contracts and report**

Private technical report: whole-family definition, landscape, composition and scoped evidence. Modern matched controls, trained sparse parity and measured hardware energy remain open.

M1 **IEA — Key Questions on Energy and AI, executive summary**

485 TWh datacenter electricity in 2025; central projection 950 TWh in 2030. Sector context, not addressable revenue.

M2 **GSMA — The Mobile Economy 2026**

5.8 billion unique mobile subscribers and 8.8 billion connections. Subscribers are not annual device sales or customers.

M3 **IFR — World Robotics 2026 release, 24 September 2026**

5 million operating industrial robots in 2025 and more than 600,000 annual installations; no adoption by this project.

T2 **Fedus et al. — Switch Transformers**

Capacity beyond selected computation has established MoE precedent; sparsity alone is not a novelty claim.

Public precedents establish relevance, not endorsement.

Historical roles are self-reported; inventorship is checked against the primary published document.

S1 AMD — Silo AI acquisition completion, 12 August 2024

Announced approximately \$665M all-cash acquisition; Poro and Viking on AMD. Mature-team strategic precedent, not a pre-seed valuation comparable.

S2 AMD — Acquisition announcement and LUMI collaboration, 10 July 2024

AMD highlighted enterprise solutions, expertise, software and Poro/Viking training on LUMI using AMD hardware; no single exclusive acquisition cause established.

S3 Liquid AI — \$250M Series A announcement, 13 December 2024

Financing announcement and AMD collaboration on efficient models; funding amount is not a disclosed company valuation.

S4 AMD Ventures — investment focus

AI models, infrastructure and silicon investment focus; published Series A-C focus means this project may be earlier than typical admission.

S5 Intel — Hala Point research system, 17 April 2024

Loihi 2 neuromorphic research prototype, event computing and prospective continuous learning. No Intel interest in this project claimed.

S6 Google — Eighth-generation TPU architecture announcement

TPU 8t/8i, model-hardware co-design and sparse MoE infrastructure. Existing internal capability creates both potential fit and competition.

F2 European Patent Office — published application EP4148389A2

Primary published document names Tero Juhani Keski-Valkama as inventor and HERE Global B.V. as applicant. Experience evidence, not Sleeping Machines-owned IP or proof of a grant.